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ABSTRACT

In this article, we consider a new $(3 + 1)$ -dimensional evolution equation, which can be used to interpret the propagation of nonlinear waves in the oceans and seas. We effectively investigate the integrable properties of the considered nonlinear evolution equation through several aspects. First of all, we present some elementary properties of multi-dimensional Bell polynomial theory and its relation with Hirota bilinear form. Utilizing those relations, we derive a Hirota bilinear form and a bilinear Bäcklund transformation. By employing the Cole–Hopf transformation in the bilinear Bäcklund transformation, we present a Lax pair. Additionally, using the Bell polynomial theory, we compute an infinite number of conservation laws. Moreover, we obtain one-, two-, and three-soliton solutions explicitly from Hirota bilinear form and illustrate them graphically. Breather solutions are also derived by employing appropriate complex conjugate parameters in the two-soliton solution. Choosing the generalized algorithm for rogue waves derived from the N-soliton solution, we directly obtain a first-order center-controllable rogue wave. Lump solutions are formulated by employing a well-established quadratic test function as a solution to the Hirota bilinear form. Further taking the test function in a combined form of quadratic and exponential functions, we obtain lump-multi-stripe solutions. Furthermore, a combined form of quadratic and hyperbolic cosine functions produces lump-multi-soliton solutions. The fission and fusion effects in the evolution of lump-multi-stripe solutions and lump-soliton-solutions are demonstrated pictorially.

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I. INTRODUCTION

In the past few decades, there has been a noticeable shift among researchers toward nonlinear models, driven by the rapid advancements in science and computer technology. Nonlinear evolution equations (NLEEs) play a pivotal role in describing a myriad of phenomena across various scientific disciplines, including physics, engineering, biology, and finance. These equations capture the complexity inherent in many natural processes, allowing for a more accurate representation of real-world dynamics like fluid mechanics,¹ optical fibers,² oceanography,³ solid-state physics,⁴ geochemistry,⁵ plasma physics,⁶ nonlinear

optics,⁷ and wave propagation in shallow water.⁸ In recent years, there has been a notable focus on the exploration of constructing exact solutions for these nonlinear evolution equations (NLEEs) to enhance our understanding of their dynamical properties with greater accuracy. Due to the growing interest in nonlinear partial differential equations (NLPDEs), researchers are actively exploring a broader range of analytical and numerical methods to study them, leading to the computation of various novel types of exact solutions.^{9–14} Studies on the integrability of nonlinear evolution equations become crucial for researchers, as it ensures the existence of exact solutions. While a

precise definition of integrability is elusive, indicators such as the Hirota bilinear form,¹⁵ inverse Scattering method,¹⁶ Painlevé analysis,^{17–19} bilinear neural network method,²⁰ bilinear Bäcklund transformation,²¹ bilinear residual network method,²² Lax pair,²³ Darboux transformation method,²⁴ Lie symmetry analysis,^{25–27} infinite conservation laws,²⁸ etc. can effectively characterize integrability. Utilizing a Lax pair for a nonlinear system, a series of integrable properties, including Hamiltonian structures²⁹ and an infinite number of conserved quantities,³⁰ can be demonstrated. Among them, the Hirota bilinear method is one of the most conventional methods and recognized as a powerful approach for analyzing NLEEs. A pivotal aspect of this method involves expressing the original equation in a bilinear form. Upon formulating the bilinear representation of nonlinear evolution equations (NLEEs), one can directly acquire quasi-periodic wave solutions, rational solutions, multi-soliton solutions, and various other exact solutions through the utilization of the bilinear structure.^{31–39} Kumar and Mohan recently proposed a systematic computational method to ascertain the bilinear form of nonlinear partial differential equations.⁴⁰ Using this method, it becomes straightforward to derive the Hirota bilinear form by employing symbolic computational software such as Mathematica, Maple, and Matlab. Lambert *et al.*^{41–43} introduced an innovative approach for obtaining the Hirota bilinear form of nonlinear evolution equations (NLEEs) through the application of Bell polynomial theory. Through this approach, the bilinear Bäcklund transformation and Lax pair can be directly derived from the Hirota bilinear form. Fan⁴⁴ expanded upon this method to directly derive infinite conservation laws from the bilinear Bäcklund transformation. Lately, numerous researchers have utilized this approach to study the integrability of diverse types of nonlinear evolution equations (NLEEs) and have yielded a variety of exact solutions.^{45–51} Researchers have shown a growing interest in the intriguing dynamical properties of exact solutions of nonlinear evolution equations (NLEEs), especially in phenomena such as solitons, lumps, breathers, and rogue waves. While waves travel through nonlinear media, variations in their amplitudes and widths are typically occur. Nevertheless, under specific conditions, the interaction between nonlinearity and dispersion can give a rise to the creation of enduring and localized waves, commonly referred to as solitons. A rogue wave also known as a giant wave is an exceptionally high wave that can manifest abruptly and vanish without a trace. These waves are localized both in space and time and were initially observed in the oceans.⁵² The phenomenon of rogue waves has been experimentally verified in nonlinear optics, generated by the generalized nonlinear Schrödinger equation.⁵³ In recent years, rogue waves have been observed in various scientific research fields such as Bose–Einstein condensates,⁵⁴ superfluids,⁵⁵ plasma physics,⁵⁶ capillary flow,⁵⁷ nonlinear optical fibers,⁵⁸ and even in finance.⁵⁹ There are only a few methods available for computing rogue waves, and among them, the Darboux transform method^{60–62} and the Hirota bilinear method^{63–65} have been gaining significant popularity. Recently, Zhaqilao⁶⁶ proposed a systematic approach for computing higher-order center-controlled rogue waves based on the N-soliton solution. Utilizing this methodology, researchers have derived higher-order center controlled rogue waves for a variety of nonlinear evolution equations.^{67–71} A breather solution is a type of partially localized solution characterized by a periodic structure in a specific direction. Breather solutions can be categorized into three

types: Akhmediev breathers, Kuznetsov–Ma breathers, and Peregrine solitons. Breather of NLEEs can be obtained by using some standard methods like the Darboux transform method,⁷² the Riemann–Hilbert approach,^{30,73} and the Hirota bilinear method.⁶⁴ Lump solution is also a rational type solution and localized in all direction. Lump solution⁷⁴ of NLEEs can be obtained directly by taking the quadratic function as a solution into the Hirota bilinear form of that equation. Numerous researchers worldwide have successfully obtained lump solutions for various nonlinear evolution equations using this method.^{75–78} Recently, Ma *et al.* obtained a new type of lump solution using the Hirota bilinear method for the generalized Calogero–Bogoyavlenskii–Schiff equation.⁷⁶ The exploration of the dynamic interplay between lump solutions and various exact solutions such as soliton, kink, breather, etc. has emerged as a highly active area of research among researchers.^{79,80} Lü and Chen⁸¹ establish a necessary and sufficient condition for obtaining lump-multi-stripe and lump-multi-soliton solution with the help of Hirota bilinear form. Utilizing these theorem, they briefly analyze the dynamical behavior of lump-multi-stripe solution and lump-multi-soliton solution of three different (2 + 1)-dimensional nonlinear evolution equations. Lü and Chen⁸² additionally establish a necessary and sufficient condition for determining lump-multi-kink solutions of nonlinear evolution equations (NLEEs) that exhibit a specific type of Hirota bilinear form. Furthermore, Mandal *et al.*⁸³ generalize the Lü’s theorem and illustrate the evolution of lump-multi-kink solution for (3 + 1)-dimensional negative order KdV–CBS equation.

Recently, Akinyemi proposed a new (2 + 1)-dimensional evolution equation⁸⁴

$$A u_{xt} + a u_{xx} + b (u^2)_{xx} + c u_{xxxx} + d u_{yy} = 0, \quad (1)$$

where a , b , and c are arbitrary real constants. Equation (1) describes the motion of water characterized by small-amplitude and long waves. Additionally, this mathematical model emphasizes few real life phenomenon such as tsunamis and tidal oscillations. Akinyemi shown that Eq. (1) is Painlevé integrable and derived various exact solutions. Wazwaz extend Eq. (1) to the following new (3 + 1)-dimensional equation:⁸⁵

$$u_{xt} + a u_{xx} + b (u^2)_{xx} + c u_{xxxx} + d u_{yy} + \alpha u_{xy} + \beta u_{xz} + \gamma u_{yz} + \delta u_{zz} = 0, \quad (2)$$

where a , b , c , d , α , β , γ , and δ are arbitrary real constants. Equation (2) finds utility in the representation and analysis of the intricate dynamics governing wave propagation in oceans and seas, offering noteworthy prospects for applications in fluid mechanics, plasma physics, and allied fields. In Ref. 85, Wazwaz shown that Eq. (2) also passes the Painlevé integrability test and studied multiple soliton solutions and lump solutions.

We observed a gap in the existing literature regarding the investigation of the bilinear Bäcklund transformation, Lax pair, and infinite conservation laws for Eq. (2). This observation motivated us to study these aspects in the current exposition. Our primary objective in this article is to explore the integrability and exact solutions of the nonlinear evolution described by Eq. (2).

The organization of our article is as follows. In Sec. II, we present some elementary properties of Bell polynomial theory and their relation with Hirota D -operator. Section III is entirely dedicated to

examining the Hirota bilinear form, bilinear Bäcklund transformation, and associated Lax pair formulation of Eq. (2) through the utilization of Bell Polynomial theory. In Sec. IV, we derive infinitely many conservation laws. In Sec. V, we calculate one-, two-, and three-soliton solution and present their evolution graphically. In Sec. VI, we derive first order breather solution. In Sec. VII, we explore first order center controlled rogue wave of Eq. (2). In Sec. VIII, we obtain lump solution of the considered equation. Sections IX and X are exclusively focused on computing the lump-multi-stripe solution and lump-multi-soliton solution as well as analyzing their dynamic behavior. In Sec. XI, we have depicted the obtained solutions and discussed their propagation and nonlinear interaction with other solutions elaborately. Finally, in Sec. XII, we draw conclusions based on our work.

II. MULTI-DIMENSIONAL BELL POLYNOMIALS

In this section, we provide a brief overview of the fundamental principles and representations of Bell polynomials.^{41,42} Assume η is a C^∞ function of t . The one-dimensional Bell polynomial, as defined in Ref. 41, is expressed as

$$Y_{nt}(\eta) = Y_n(\eta_1, \eta_2, \dots, \eta_{nt}) = e^{-\eta} \partial_t^n e^\eta, \quad n = 1, 2, 3, \dots \quad (3)$$

Several one-dimensional Bell polynomials can be derived from the aforementioned statement as

$$Y_t = \eta_t, \quad Y_{2t} = \eta_{2t} + \eta_t^2, \quad Y_{3t} = \eta_{3t} + 3\eta_t \eta_{2t} + \eta_t^3, \dots \quad (4)$$

We use the formula

$$Y_{nt}(\eta) = \sum \frac{n!}{a_1! a_2! \dots a_n!} \left(\frac{\eta_t}{1!} \right)^{a_1} \left(\frac{\eta_{2t}}{2!} \right)^{a_2} \dots \left(\frac{\eta_{nt}}{n!} \right)^{a_n}, \quad (5)$$

where the sum runs over all partitions of $n = a_1 + 2a_2 + \dots + na_n$ and obtains the aforementioned expressions, Eq. (4). Assuming $\eta = \eta(t_1, t_2, \dots, t_s)$ as a C^∞ multi-variable function, we can extend the dimension of the Bell polynomial. The multi-dimensional Bell polynomial can then be expressed as follows:

$$Y_{n_1 t_1, \dots, n_s t_s}(\eta) \equiv Y_{n_1, \dots, n_s}(\eta_{m_1 t_1, \dots, m_s t_s}) = e^{-\eta} \partial_{t_1}^{n_1} \dots \partial_{t_s}^{n_s} e^\eta, \quad (6)$$

where $\eta_{m_1 t_1, \dots, m_s t_s} = \partial_{t_1}^{m_1} \dots \partial_{t_s}^{m_s} \eta$, $m_i = 0, 1, \dots, n_i$, and $i = 1, 2, \dots, s$. Here, $Y_{n_1 t_1, \dots, n_s t_s}(\eta)$ denotes the multi-variable Bell polynomial with respect to $\eta_{m_1 t_1, \dots, m_s t_s}$. Specifically, if we select η as a function of both t and z , the corresponding lower-order two-dimensional Bell polynomials can be derived as follows:

$$Y_{2t}(\eta) = \eta_{2t} + \eta_t^2, \quad Y_{3t}(\eta) = \eta_{3t} + 3\eta_t \eta_{2t} + \eta_t^3, \quad (7)$$

$$Y_{t,z} = \eta_{t,z} + \eta_t \eta_z, \quad Y_{2t,z}(\eta) = \eta_{2t,z} + \eta_{2t} \eta_z + 2\eta_{t,z} \eta_t + \eta_t^2 \eta_z, \dots \quad (8)$$

According to the aforementioned one-dimensional Bell polynomials, Eq. (6), we can depict multi-dimensional binary Bell polynomials as follows:

$$\mathcal{Y}_{n_1 t_1, \dots, n_s t_s}(\mathcal{f}, \mathcal{g}) = Y_{n_1 t_1, \dots, n_s t_s}(\eta), \quad (9)$$

where

$$\eta_{m_1 t_1, \dots, m_s t_s} = \begin{cases} \mathcal{f}_{m_1 t_1, \dots, m_s t_s}, & m_1 + \dots + m_s \text{ is odd,} \\ \mathcal{g}_{m_1 t_1, \dots, m_s t_s}, & m_1 + \dots + m_s \text{ is even.} \end{cases} \quad (10)$$

The following are a few two-dimensional binary Bell polynomials that can be derived from the above-mentioned statement:

$$\begin{aligned} \mathcal{Y}_t(\mathcal{f}) &= \mathcal{f}_t, \quad \mathcal{Y}_{2t}(\mathcal{f}, \mathcal{g}) = \mathcal{g}_{2t} + \mathcal{f}_t^2, \\ \mathcal{Y}_{t,z}(\mathcal{f}, \mathcal{g}) &= \mathcal{g}_{t,z} + \mathcal{f}_t \mathcal{f}_z, \\ \mathcal{Y}_{2t,z}(\mathcal{f}, \mathcal{g}) &= \mathcal{f}_{2t,z} + \mathcal{g}_{2t} \mathcal{f}_z + 2\mathcal{g}_{t,z} \mathcal{f}_t + \mathcal{f}_t^2 \mathcal{f}_z, \\ \mathcal{Y}_{3t} &= \mathcal{f}_{3t} + 3\mathcal{g}_{2t} \mathcal{f}_t + \mathcal{f}_t^3, \dots \end{aligned} \quad (11)$$

With the help of the identity

$$(\eta\psi)^{-1} D_{t_1}^{n_1} \dots D_{t_s}^{n_s} \eta \cdot \psi = \mathcal{Y}_{n_1 t_1, \dots, n_s t_s}(\mathcal{f} = \ln \eta / \psi, \mathcal{g} = \ln \eta \psi), \quad (12)$$

we can establish a relation between the conventional Hirota bilinear expression $D_{t_1}^{n_1} \dots D_{t_s}^{n_s} \eta \cdot \psi$ and binary Bell polynomials, where the D -operator is presented by Hirota¹⁵ as follows:

$$\begin{aligned} D_{t_1}^{n_1} \dots D_{t_s}^{n_s} \eta \cdot \psi &= (\partial_{t_1} - \partial_{t'_1})^{n_1} \dots (\partial_{t_s} - \partial_{t'_s})^{n_s} \eta(t_1, \dots, t_s) \\ &\quad \cdot \psi(t'_1, \dots, t'_s) \Big|_{t'_1=t_1, \dots, t'_s=t_s}. \end{aligned}$$

In case when $\eta = \psi$, the identity Eq. (12) becomes

$$\begin{aligned} (\eta)^{-2} D_{t_1}^{n_1} \dots D_{t_s}^{n_s} \eta \cdot \eta &= \mathcal{Y}_{n_1 t_1, \dots, n_s t_s}(\mathcal{f} = 0, \mathcal{g} = 2 \ln \eta) \\ &= \begin{cases} 0, & n_1 + \dots + n_s \text{ is odd,} \\ \mathcal{P}_{n_1 t_1, \dots, n_s t_s}(\mathcal{p}), & n_1 + \dots + n_s \text{ is even.} \end{cases} \end{aligned} \quad (13)$$

Here, the $\mathcal{P}$ -polynomials refer to the even-ordered $\mathcal{Y}$ -polynomials, and a few of the initial ones are listed as follows:

$$\begin{aligned} \mathcal{P}_{2t}(\mathcal{p}) &= \mathcal{p}_{2t}, \quad \mathcal{P}_{t,z}(\mathcal{p}) = \mathcal{p}_{t,z}, \\ \mathcal{P}_{3t,z}(\mathcal{p}) &= \mathcal{p}_{3t,z} + 3\mathcal{p}_{t,z} \mathcal{p}_{2t}, \quad \mathcal{P}_{4t}(\mathcal{p}) = \mathcal{p}_{4t} + 3\mathcal{p}_{2t}^2. \end{aligned} \quad (14)$$

The binary Bell polynomial $\mathcal{Y}_{n_1 t_1, \dots, n_s t_s}(\mathcal{f}, \mathcal{g})$ can be written as a linear combination of $\mathcal{P}$ -polynomials and Bell polynomials $Y_{n_1 t_1, \dots, n_s t_s}(\mathcal{f})$ as

$$(\eta\psi)^{-1} D_{t_1}^{n_1} \dots D_{t_s}^{n_s} \eta \cdot \psi = \mathcal{Y}_{n_1 t_1, \dots, n_s t_s}(\mathcal{f}, \mathcal{g}),$$

where

$$\mathcal{f} = \ln \eta / \psi \quad \text{and} \quad \mathcal{g} = \ln \eta \psi = \mathcal{Y}_{n_1 t_1, \dots, n_s t_s}(\mathcal{f}, \mathcal{f} + \mathcal{p}),$$

where

$$\begin{aligned} \mathcal{f} &= \ln \eta / \psi \quad \text{and} \quad \mathcal{p} = 2 \ln \psi \\ &= \sum_{m_1=0}^{n_1} \dots \sum_{m_s=0}^{n_s} \prod_{i=1}^s \binom{n_i}{m_i} \mathcal{P}_{m_1 t_1, \dots, m_s t_s}(\mathcal{p}) Y_{(n_1-m_1)t_1, \dots, (n_s-m_s)t_s}(\mathcal{f}). \end{aligned} \quad (15)$$

Using Hopf-Cole transformation $\mathcal{f} = \ln \psi$, binary Bell polynomial can be expressed in the following form:

$$Y_{n_1 t_1, \dots, n_s t_s}(\mathcal{f} = \ln \psi) = \frac{\psi_{n_1 t_1, \dots, n_s t_s}}{\psi}, \quad (16)$$

through which, Eq. (15) can be reexpressed as

$$\begin{aligned} (\eta\psi)^{-1} D_{t_1}^{n_1} \dots D_{t_s}^{n_s} \eta \cdot \psi &= \psi^{-1} \sum_{m_1=0}^{n_1} \dots \sum_{m_s=0}^{n_s} \prod_{i=1}^s \binom{n_i}{m_i} \mathcal{P}_{m_1 t_1, \dots, m_s t_s}(\mathcal{p}) \psi_{(n_1-m_1)t_1, \dots, (n_s-m_s)t_s}. \end{aligned} \quad (17)$$

Equation (17) provides the most straightforward approach to constructing the associated Lax pair for the relevant nonlinear evolution equation. The Bell polynomial theory concepts mentioned earlier will be further employed to establish the bilinear form, bilinear Bäcklund transformation, and the Lax pair formulation of Eq. (2).

III. HIROTA BILINEAR FORM, BILINEAR BÄCKLUND TRANSFORMATION, AND LAX PAIR

In order to achieve Hirota bilinear form of Eq. (2), we introduce a new potential field q by setting

$$u = \frac{3c}{b} q_{xx}. \quad (18)$$

Substituting Eq. (18) in Eq. (2) and integrate twice it with respect to x yields

$$E(q) = q_{xt} + a q_{2x} + c(q_{4x} + 3q_{2x}^2) + d q_{2y} + \alpha q_{xy} + \beta q_{xz} + \gamma q_{yz} + \delta q_{2z} = 0. \quad (19)$$

Furthermore, we choose $q = 2 \log f$, and we deduce the Hirota bilinear of Eq. (2) by using the relation between P-polynomial and Hirota D -operator as

$$(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2) f \cdot f = 0. \quad (20)$$

To formulate bilinear Bäcklund transformation, we choose $q' = 2 \log f'$ as a distinct solution of Eq. (2). Furthermore, we introduce two new variables $v = \frac{q' - q}{2}$ and $w = \frac{q' + q}{2}$. Then, the corresponding two-field condition can be written as

$$\begin{aligned} E(q') - E(q) &= 2[v_{xt} + a v_{2x} + c(v_{4x} + 6v_{2x} w_{2x}) + d v_{2y} \\ &\quad + \alpha v_{xy} + \beta v_{xz} + \gamma v_{yz} + \delta v_{2z}] \\ &= 2 \frac{\partial}{\partial x} [\mathcal{Y}_t(v) + c \mathcal{Y}_{3x}(vw) + \beta \mathcal{Y}_z(v)] + 2 \mathcal{R}(v, w), \end{aligned} \quad (21)$$

where $\mathcal{R}(v, w) = a v_{2x} + d v_{2y} + \alpha v_{xy} + \gamma v_{yz} + \delta v_{2z} + 3c$ Wronskian $[\mathcal{Y}_{2x}(v, w), \mathcal{Y}_x(v)]$. In order to write $\mathcal{R}(v, w)$ in the expression of $\mathcal{Y}$ polynomials, we introduce a constraint as follows:

$$\mathcal{Y}_{2x}(v, w) + A \mathcal{Y}_z(v, w) = \eta, \quad (22)$$

where A is an undetermined constant and η is an arbitrary parameter.

Taking $3cA^2 = \delta$, $\mathcal{R}(v, w)$ can be written as

$$\begin{aligned} \mathcal{R}(v, w) &= \frac{\partial}{\partial x} [(3c\eta + a) \mathcal{Y}_x(v) - 3cA \mathcal{Y}_{xz}(v, w)] \\ &\quad + \frac{\partial}{\partial y} [\alpha \mathcal{Y}_x(v) + \gamma \mathcal{Y}_z(v) + d \mathcal{Y}_y(v)]. \end{aligned} \quad (23)$$

Finally, Eq. (21) can be written as

$$\begin{aligned} E(q') - E(q) &= 2 \frac{\partial}{\partial x} [\mathcal{Y}_t(v) + c \mathcal{Y}_{3x}(v, w) + \beta \mathcal{Y}_z(v) \\ &\quad + (3c\eta + a) \mathcal{Y}_x(v) - 3cA \mathcal{Y}_{xz}(v, w)] \\ &\quad + 2 \frac{\partial}{\partial y} [\alpha \mathcal{Y}_x(v) + d \mathcal{Y}_y(v) + \gamma \mathcal{Y}_z(v)]. \end{aligned} \quad (24)$$

Decoupling Eq. (24), we obtain the bilinear Bäcklund transformation of Eq. (2) as follows:

$$\mathcal{Y}_{2x}(v, w) + A \mathcal{Y}_z(v, w) = \eta, \quad (25a)$$

$$\mathcal{Y}_t(v) + c \mathcal{Y}_{3x}(v, w) + \beta \mathcal{Y}_z(v) + (3c\eta + a) \mathcal{Y}_x(v) - 3cA \mathcal{Y}_{xz}(v, w) = \eta_1, \quad (25b)$$

$$\alpha \mathcal{Y}_x(v) + d \mathcal{Y}_y(v) + \gamma \mathcal{Y}_z(v) = \eta_2. \quad (25c)$$

With the help of Eq. (12), the aforesaid bilinear Bäcklund transformation Eq. (25) also can be written in terms of Hirota D -operator as follows:

$$(f' \cdot f)^{-1} [D_x^2 + A D_z - \eta] (f' \cdot f) = 0, \quad (26a)$$

$$(f' \cdot f)^{-1} [D_t + c D_x^3 + \beta D_z + (3c\eta + a) D_x - 3cA D_x D_z - \eta_1] (f' \cdot f) = 0, \quad (26b)$$

$$(f' \cdot f)^{-1} [\alpha D_x + d D_y + \gamma D_z - \eta_2] (f' \cdot f) = 0. \quad (26c)$$

Using the Cole-Hopf transformation $v = \log \zeta$ and linearizing the Bell polynomial system, Eq. (25), we derive the Lax pair of Eq. (2) as

$$\zeta_{2x} + A \zeta_z + (q_{2x} - \eta) \zeta = 0, \quad (27a)$$

$$\begin{aligned} \zeta_t + \beta \zeta_z + c(\zeta_{xz} + \zeta_{3x} - 3A) + (a + 3c\eta + 3c q_{2x}) \zeta_x \\ - 3cA q_{xz} \zeta - \eta_1 \zeta = 0, \end{aligned} \quad (27b)$$

$$\alpha \zeta_x + d \zeta_y + \gamma \zeta_z - \eta_2 \zeta = 0. \quad (27c)$$

IV. INFINITELY MANY CONSERVATION LAWS

To construct infinite conservation laws of Eq. (2), we rewrite Eqs. (22) and (24) in following form:

$$v_x^2 + A v_z - \eta = 0, \quad (28)$$

$$\begin{aligned} \frac{\partial}{\partial t} (v_x) + \frac{\partial}{\partial x} [c(v_{3x} + 3v_x w_{2x} + v_x^3) + (3c\eta + a) v_x - 3cA v_x v_z] \\ + \frac{\partial}{\partial y} (\alpha v_x + d v_y + \gamma v_z) + \frac{\partial}{\partial z} (-3cA \eta + \delta v_z + 3cA v_x^2 + \beta v_z) = 0. \end{aligned} \quad (29)$$

We present a novel potential function $\xi = \frac{q' - q_x}{2}$, which yields

$$v_x = \xi, \quad v_y = \partial_x^{-1}(\xi_y), \quad v_z = \partial_x^{-1}(\xi_z), \quad w_x = q_x + \xi. \quad (30)$$

Substitution of Eq. (30) into Eq. (28) and Eq. (29) yields a Riccati type equation and a divergence type equation as

$$\xi^2 + q_{2x} + \xi_x + A \partial_x^{-1}(\xi_z) - \eta = 0, \quad (31)$$

$$\begin{aligned} \xi_t + \partial_x [c \xi_{2x} + 6c \xi \eta - 2c \xi^3 - 6cA \xi \partial_x^{-1}(\xi_z) + a \xi] \\ + \partial_z [-3cA \eta + \delta \partial_x^{-1} \xi_z + 3cA \xi^2 + \beta \partial_x^{-1}(\xi_z)] \\ + \partial_y [\alpha \xi + d \partial_x^{-1}(\xi_y) + \gamma \partial_x^{-1}(\xi_z)] = 0. \end{aligned} \quad (32)$$

Setting $\eta = \xi^2$ and $\xi = \xi' + \epsilon$ in Eq. (31), we obtain

$$q_{2x} + \xi_x' + 2\xi' \epsilon + \xi'^2 + A \partial_x^{-1}(\xi_z') - \epsilon^2 = 0. \quad (33)$$

We choose infinite series form of ξ' as

$$\xi' = \sum_{n=1}^{\infty} K_n(q, q_x, q_{2x}, \dots) \epsilon^{-n}. \quad (34)$$

Substituting Eq. (34) into Eq. (33) and equating all the like powers of ϵ , finally, we derive the recursions relations for the conserved densities as

$$K_1 = -\frac{1}{2} \mathcal{Q}_{2x}, \quad (35a)$$

$$K_2 = -\frac{1}{2} [K_{1,x} + A \partial_x^{-1} (K_{1,z})] = \frac{1}{4} (\mathcal{Q}_{3x} + A \mathcal{Q}_{x,z}), \quad (35b)$$

$$K_3 = -\frac{1}{2} [K_{2,x} + A \partial_x^{-1} (K_{2,z}) + K_1^2] = -\frac{1}{8} [\mathcal{Q}_{4x} + 2A(\mathcal{Q}_{2xz} + \mathcal{Q}_{2z}) \mathcal{Q}_{2x}^2], \quad (35c)$$

$$K_{n+1} = -\frac{1}{2} \left(K_{n,x} + A \partial_x^{-1} K_{n,z} + \sum_{i=1}^{n-1} K_i K_{n-i} \right). \quad (35d)$$

Again substituting Eq. (34) with $\eta = \xi^2$ and $\xi = \xi' + \epsilon$ into Eq. (32), we have

$$\begin{aligned} & \sum_{n=1}^{\infty} K_{n,t} \epsilon^{-n} + \partial_x \left[c \sum_{n=1}^{\infty} K_{n,2x} \epsilon^{-n} + 4c\epsilon^3 - 2c \left(\sum_{n=1}^{\infty} K_n \epsilon^{-n} \right)^3 \right. \\ & - 6c\epsilon \left(\sum_{n=1}^{\infty} K_n \epsilon^{-n} \right)^2 - 6cA \left(\sum_{n=1}^{\infty} K_n \epsilon^{-n} \right) \partial_x^{-1} \left(\sum_{n=1}^{\infty} K_{n,z} \epsilon^{-n} \right) \\ & + a \left(\sum_{n=1}^{\infty} K_n \epsilon^{-n} + \epsilon \right) \left. \right] + \partial_y \left[\alpha \left(\epsilon + \sum_{n=1}^{\infty} K_n \epsilon^{-n} \right) \right. \\ & - 6c\epsilon + d \partial_x^{-1} \left(\sum_{n=1}^{\infty} K_{n,y} \epsilon^{-n} \right) + \gamma \partial_x^{-1} \left(\sum_{n=1}^{\infty} K_{n,z} \epsilon^{-n} \right) \left. \right] \\ & + \partial_z \left[3cA \left(\sum_{n=1}^{\infty} K_n \epsilon^{-n} \right)^2 + 6cA \left(\sum_{n=1}^{\infty} K_n \epsilon^{-n} \right) \epsilon \right. \\ & \left. + (\delta + \beta) \partial_x^{-1} \left(\sum_{n=1}^{\infty} K_{n,z} \epsilon^{-n} \right) \right] = 0. \quad (36) \end{aligned}$$

Comparing the coefficient of all the like powers of ϵ from Eq. (36), we obtain the conservation laws of Eq. (2) as

$$K_{n,t} + \mathcal{L}_{n,x} + \mathcal{M}_{n,y} + \mathcal{N}_{n,z} = 0, \quad (37)$$

where

$$\mathcal{L}_1 = cK_{1,2x} + aK_1 - 6cK_1^2 - 6cA \partial_x^{-1} K_{2,z}, \quad (38a)$$

$$\mathcal{L}_2 = cK_{2,2x} + aK_2 - 12cK_1K_2 - 6cA(\partial_x^{-1}K_{3,z} + K_1\partial_x^{-1}K_{1,z}), \dots \dots \quad (38b)$$

$$\begin{aligned} \mathcal{L}_n = & aK_n + c \left(K_{n,2x} - 6 \sum_{i=1}^n K_i K_{n-i+1} - 2 \sum_{i_1+i_2+i_3=n} K_{i_1} K_{i_2} K_{i_3} \right) \\ & - 6cA \left(\partial_x^{-1} K_{n+1,z} + \sum_{n=1}^n K_i \partial_x^{-1} K_{n-i,y} \right), \quad (38c) \end{aligned}$$

$$\mathcal{M}_1 = \alpha K_1 + d \partial_x^{-1} K_{1,y} + \gamma \partial_x^{-1} K_{1,z}, \quad (39a)$$

$$\mathcal{M}_2 = \alpha K_2 + d \partial_x^{-1} K_{2,y} + \gamma \partial_x^{-1} K_{2,z}, \quad (39b)$$

$$\mathcal{M}_n = \alpha K_n + d \partial_x^{-1} K_{n,y} + \gamma \partial_x^{-1} K_{n,z}, \quad (39c)$$

and

$$\mathcal{N}_1 = 6cAK_2 + (\beta + \delta)K_{1,z}, \quad (40a)$$

$$\mathcal{N}_2 = 6cAK_3 + 3cAK_1^2 + (\beta + \delta)K_{2,z}, \quad (40b)$$

$$\mathcal{N}_n = 6cAK_{n+1} + 3cA \sum_{i=1}^{n-1} K_i K_{n-i} + (\beta + \delta) \partial_x^{-1} K_{n,z}. \quad (40c)$$

The recursion relations Eq. (35) provide the values of K_n .

V. SOLITON SOLUTIONS

A. One-soliton solution

To formulate one soliton solution of Eq. (2), we assume f in the following form:

$$f = 1 + e^{\Upsilon_1}, \quad (41)$$

where $\Upsilon_1 = k_1(x + p_1y + q_1z + w_1t) + \Upsilon_1^0$ and $k_1, p_1, q_1, \Upsilon_1^0$ are arbitrary constants. Substituting Eq. (41) into Eq. (20) and equating all exponential functions to zero, we obtain dispersion relation as

$$w_1 = -(a + ck_1^2 + dp_1^2 + \alpha p_1 + \beta q_1 + \gamma p_1q_1 + \delta q_1^2). \quad (42)$$

Finally, we obtain one-soliton solution of Eq. (2) by substituting Eq. (41) along with Eq. (42) into Eq. (20) as

$$u = \frac{6c}{b} [\log(1 + e^{\Upsilon_1})]_{2x}. \quad (43)$$

In Fig. 1, we present 3D plot, density plot, and 2D plot of one soliton solution Eq. (43), corresponding to parametric values as $a = c = d = \alpha = \beta = \gamma = \delta = 1$, $k_1 = 0.4$, $p_1 = 1.2$, $q_1 = 1.3$, $\Upsilon_1^0 = 0$.

B. Two-soliton solution

In order to obtain two soliton solution of Eq. (2), we choose f in the following form:

$$f = 1 + e^{\Upsilon_1} + e^{\Upsilon_2} + B_{12}e^{\Upsilon_1+\Upsilon_2}, \quad (44)$$

where $\Upsilon_i = k_i(x + p_iy + q_iz + w_it) + \Upsilon_i^0$, ($i = 1, 2$) and $k_i, p_i, q_i, \Upsilon_i^0$, ($i = 1, 2$) are arbitrary constants. Substituting Eq. (44) into Eq. (20), we derive dispersion relation and B_{12} as

$$w_i = -(a + ck_i^2 + dp_i^2 + \alpha p_i + \beta q_i + \gamma p_iq_i + \delta q_i^2), \quad (i = 1, 2), \quad (45a)$$

$$B_{12} = \left[\frac{(k_1 - k_2)(k_1w_1 - k_2w_2) + (k_1 - k_2)^2 + (k_1 - k_2)^4 + (k_1p_1 - k_2p_2)^2 + (k_1 - k_2)(k_1p_1 + k_1q_1 - k_2p_2 - k_2q_2)}{(k_1 + k_2)(k_1w_1 + k_2w_2) + (k_1 + k_2)^2 + (k_1 + k_2)^4 + (k_1p_1 + k_2p_2)^2 + (k_1 + k_2)(k_1p_1 + k_1q_1 + k_2p_2 + k_2q_2)} \right]. \quad (45b)$$

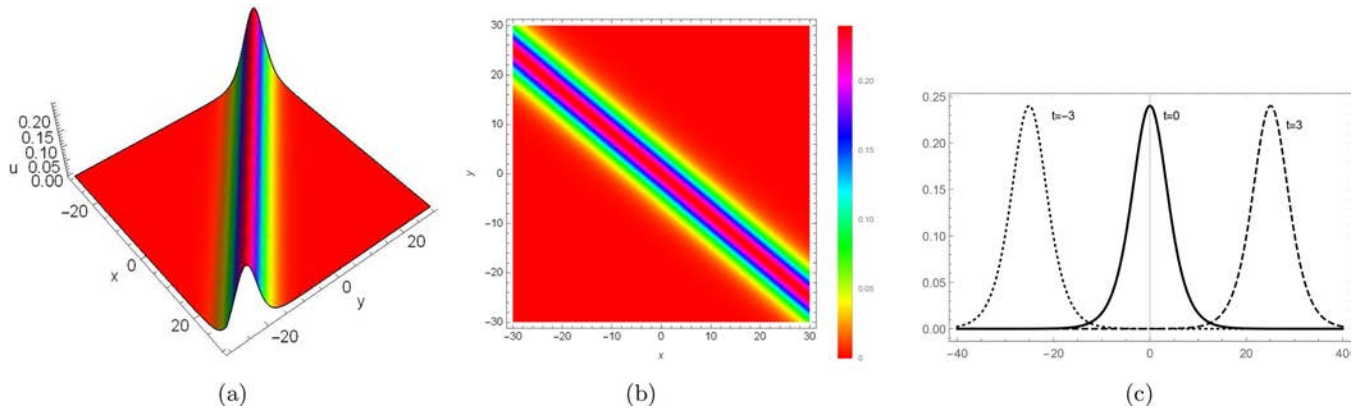


FIG. 1. One soliton solution of Eq. (2) given by Eq. (43). (a) 3D plot, (b) density plot, and (c) 2D plot at different times.

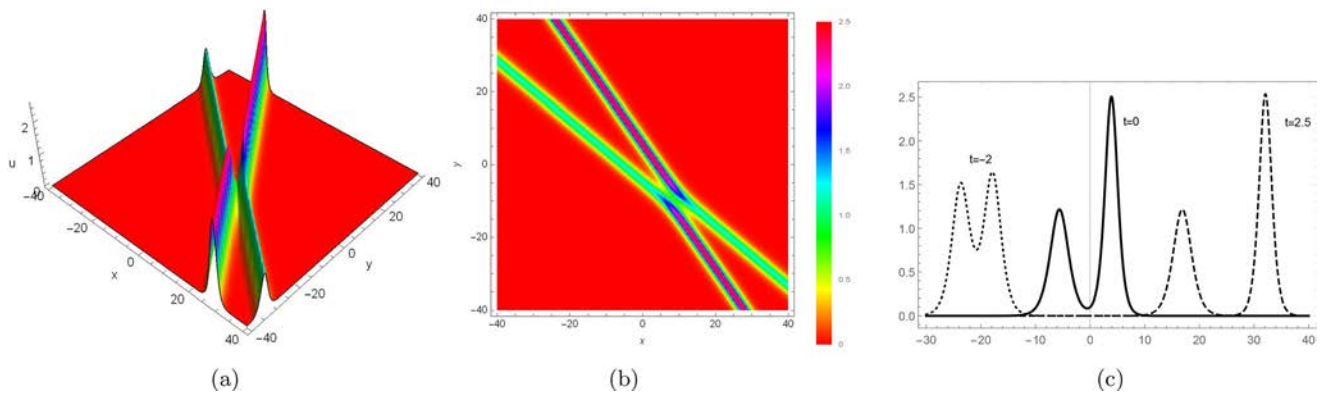


FIG. 2. Two soliton solution of Eq. (2) given by Eq. (46). (a) 3D plot, (b) density plot, and (c) 2D plot at different times.

Further substitution of Eq. (44) with Eq. (45) into Eq. (20) yields two soliton solution of Eq. (2) as

$$u = \frac{6c}{b} [\log(1 + e^{Y_1} + e^{Y_2} + B_{12}e^{Y_1+Y_2})]_{2x}. \quad (46)$$

In Fig. 2, we demonstrate 3D plot, density plot, and 2D plot of two soliton solution Eq. (46), corresponding to parametric values as $a = c = d = \alpha = \beta = \gamma = \delta = 1$, $k_1 = 0.9$, $p_1 = 1.2$, $q_1 = 1.3$, $k_2 = -1.3$, $p_2 = 0.7$, $q_2 = 2$, and $Y_i^0 = 0$ ($i = 1, 2$).

C. Three-soliton solution

In similar way, we can derive three soliton solution of Eq. (2) by choosing f in the following form:

$$f = 1 + e^{Y_1} + e^{Y_2} + e^{Y_3} + B_{12}e^{Y_1+Y_2} + B_{13}e^{Y_1+Y_3} + B_{23}e^{Y_2+Y_3} + B_{123}e^{Y_1+Y_2+Y_3}, \quad (47)$$

where $Y_i = k_i(x + p_i y + q_i z + w_i t) + Y_i^0$, ($i = 1, 2, 3$) and k_i, p_i, q_i, Y_i^0 , ($i = 1, 2, 3$) are arbitrary constants. Substituting Eq. (47) into Eq. (20), we derive dispersion relation and B_{ij} as

$$w_i = -(a + ck_i^2 + dp_i^2 + \alpha p_i + \beta q_i + \gamma p_i q_i + \delta q_i^2), \quad (i = 1, 2, 3), \quad (48a)$$

$$B_{ij} = \frac{\left[(k_i - k_j)(k_i w_i - k_j w_j) + (k_i - k_j)^2 + (k_i - k_j)^4 + (k_i p_i - k_j p_j)^2 + (k_i - k_j)(k_i p_i + k_i q_i - k_j p_j - k_j q_j) + (k_i p_i - k_j p_j)(k_i q_i - k_j q_j) + (k_i q_i - k_j q_j)^2 \right]}{\left[(k_i + k_j)(k_i w_i + k_j w_j) + (k_i + k_j)^2 + (k_i + k_j)^4 + (k_i p_i + k_j p_j)^2 + (k_i + k_j)(k_i p_i + k_i q_i + k_j p_j + k_j q_j) + (k_i p_i + k_j p_j)(k_i q_i + k_j q_j) + (k_i q_i + k_j q_j)^2 \right]}. \quad (48b)$$

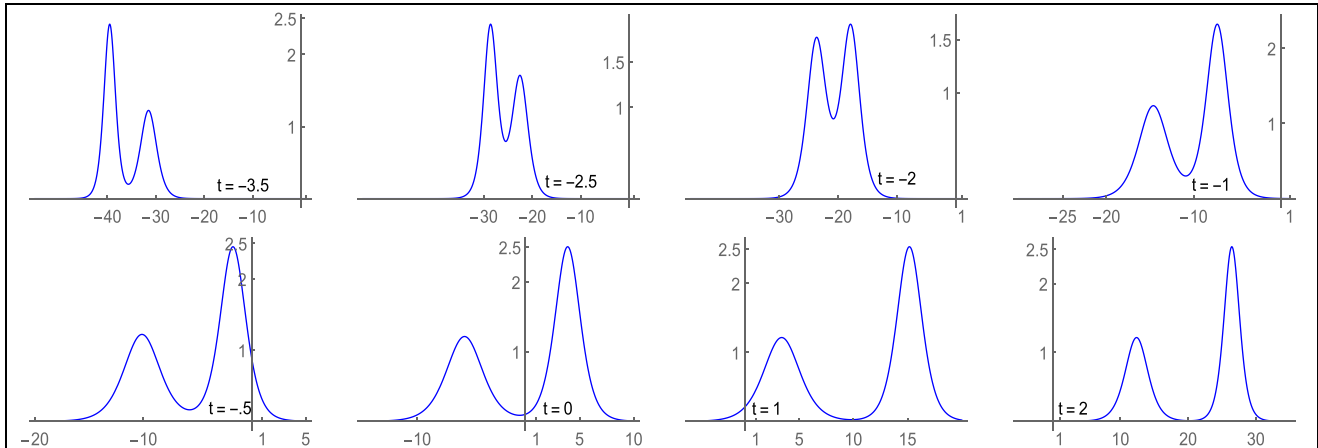


FIG. 3. Time evolution of two soliton solution of Eq. (2) for the same parameters as in Fig. 2.

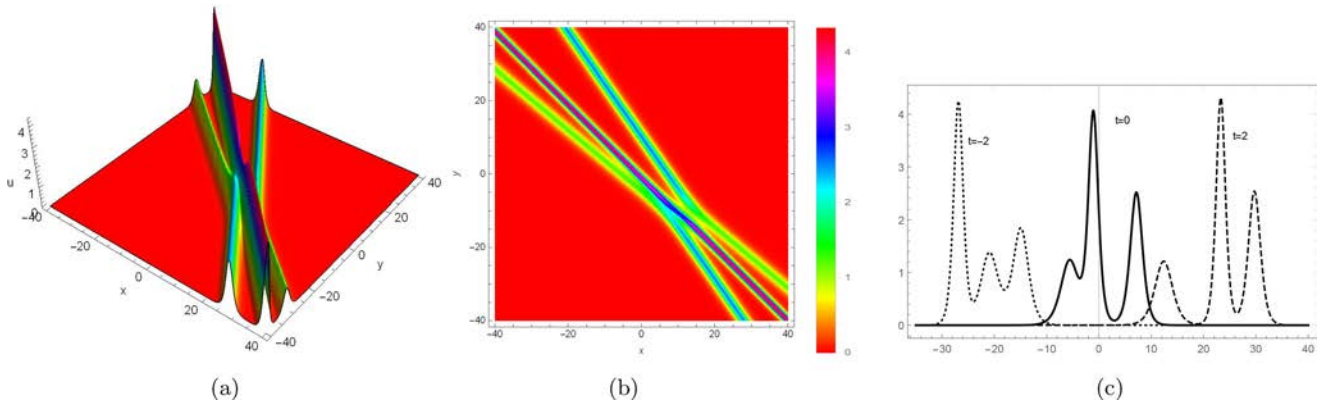


FIG. 4. Three soliton solution of Eq. (2) given by Eq. (49). (a) 3D plot, (b) density plot, and (c) 2D plot at different times.

Finally, substituting Eq. (47) with Eq. (48) into Eq. (20), we derive three soliton solution of Eq. (2) as

$$u = \frac{6c}{b} \left[\log(1 + e^{Y_1} + e^{Y_2} + e^{Y_3} + B_{12}e^{Y_1+Y_2} + B_{13}e^{Y_1+Y_3} + B_{23}e^{Y_2+Y_3} + B_{123}e^{Y_1+Y_2+Y_3}) \right]_{2x} \quad (49)$$

In Fig. 4, we illustrate three soliton solution graphically corresponding to the parametric values as $a = c = d = \alpha = \beta = \gamma = \delta = 1$, $k_1 = 0.9$, $p_1 = 1.2$, $q_1 = 1.3$, $k_2 = -1.3$, $p_2 = 0.7$, $q_2 = 2$, $k_3 = 1.7$, $p_3 = 1$, $q_3 = 1.7$, and $Y_i^0 = 0$ ($i = 1, 2, 3$).

VI. BREATHER SOLUTION

Utilizing the criteria outlined in prior studies^{86,87} for acquiring breather solutions, one can derive an analytical expression for breather solutions by selecting appropriate parameters within the framework of the two-soliton solution. In order to derive the first-order breather solution, it is necessary to express the parameters in Eq. (44) in the following complex conjugate form:

$$k_1 = k_2^* = a + ib, \quad p_1 = p_2^* = c + id, \quad q_1 = q_2^* = e + if. \quad (50)$$

In particular, taking $k_1 = k_2^* = i$, $p_1 = p_2^* = 1$, $q_1 = q_2^* = 2 + i$, and $a = b = c = d = \alpha = \beta = \gamma = \delta = 1$, Eq. (44) can be written as

$$f = 1 + 2 \cos(x + 2y + z + 6t) (\cosh(-2z - 10t) + \sinh(-2z - 10t)) - 2 \cosh(-2z - 12t) - 2 \sinh(-2z - 12t). \quad (51)$$

Substituting Eq. (51) into Eq. (20), we can obtain the first order breather solution of Eq. (2) as

$$u = 6 \left[\log(1 + 2 \cos(x + 2y + z + 6t) (\cosh(-2z - 10t) + \sinh(-2z - 10t)) - 2 \cosh(-2z - 12t) - 2 \sinh(-2z - 12t)) \right]_{xx} \quad (52)$$

VII. ROGUE WAVE SOLUTION

In order to derive rogue wave solution of Eq. (2), we use the transformation $\rho = x + y - t$, which converts Eq. (2) into

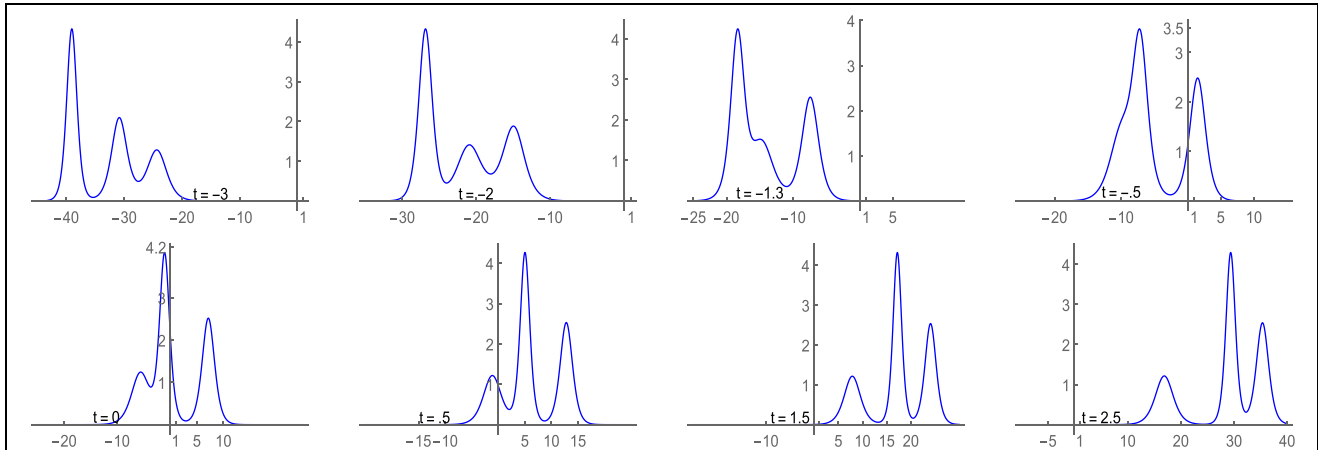
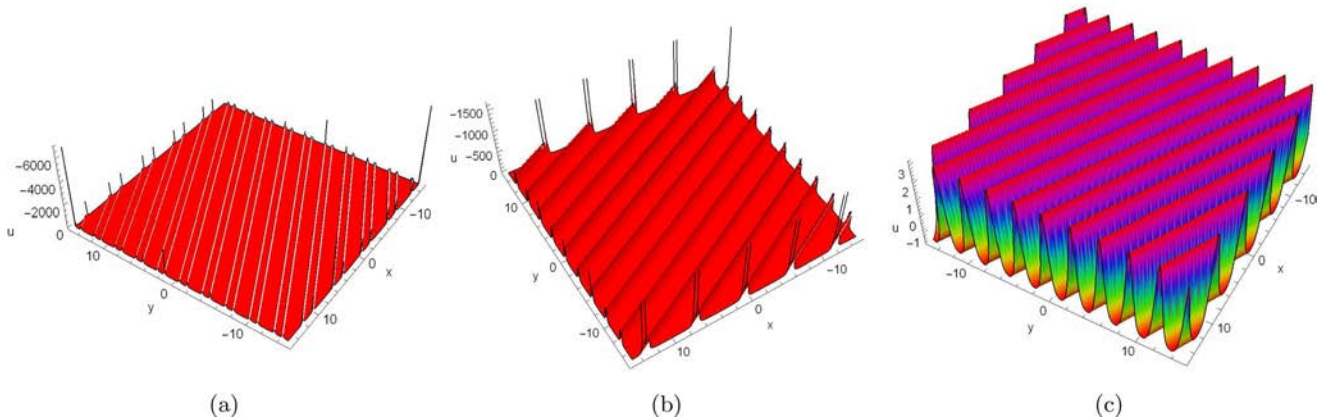


FIG. 5. Time evolution of three soliton solution of Eq. (2) for the same parameters as in Fig. 4.


 FIG. 6. Evolution of first order breather solution of Eq. (2) given by Eq. (52) at (a) $t = 0$, (b) $t = 0.165$, and (c) $t = 0.3$.

$$(a + d + \alpha - 1)u_{2\rho} + b(u^2)_{2\rho} + cu_{4\rho} + (\beta + \gamma)u_{\rho z} + u_{2z} = 0. \quad (53)$$

Using the depending variable transformation

$$u(\rho, z) = \frac{3c}{b} [2 \log g]_{\rho\rho}, \quad (54)$$

along with $c = -1$ and integrating twice with respect to ρ , we obtain Hirota bilinear equation corresponding to Eq. (53) as

$$\begin{aligned} & [(a + d + \alpha - 1)D_\rho^2 - D_\rho^4 + (\beta + \gamma)D_\rho D_z + \delta D_z^2]g \cdot g \\ &= (a + d + \alpha - 1)(g g_{2\rho} - g_\rho^2) - (g_{4\rho} g - 4g_\rho g_{2\rho} + 3g_{2\rho}^2) \\ &+ (\beta + \gamma)(g g_{\rho z} - g_\rho g_z) + \delta(g g_{2z} - g_z^2) = 0. \end{aligned} \quad (55)$$

The generalized form of center controlled rogue wave solution⁶⁷ can be considered as

$$\begin{aligned} g(\rho, z) &= \tilde{g}_n(\rho, z, c_1, c_2) \\ &= \sum_{j=0}^{\frac{n(n+1)}{2}} \sum_{i=0}^j b_{n(n+1)-2j, 2i} (z - c_2)^{2i} (\rho - c_1)^{n(n+1)-2j}, \end{aligned} \quad (56)$$

where $b_{p,q}$ ($p, q = 0, 2, 4, \dots, j(j+1)$) are constants that will be found later and c_1 and c_2 are real center controlling parameters of rogue wave.

For first order rogue wave solution, the auxiliary function $g(\rho, z)$ can be considered in the following form:

$$g(\rho, z) = b_{0,0} + b_{0,2}z^2 + b_{2,0}\rho^2. \quad (57)$$

By substituting Eq. (57) into Eq. (55) and equating the coefficient of different powers of $\rho^i z^j$ to zero, we obtain a system of equations as

$$2\alpha b_{2,0} b_{0,2} - 2\delta b_{0,2}^2 + 2d b_{2,0} b_{0,2} + 2a b_{2,0} b_{0,2} - 2b_{2,0} b_{0,2} = 0, \quad (58a)$$

$$-2\alpha b_{2,0}^2 - 2d b_{2,0}^2 + 2\delta b_{0,2} b_{2,0} - 2a b_{2,0}^2 + 2b_{2,0}^2 = 0, \quad (58b)$$

$$2b_{0,0} b_{2,0} (a + \alpha + d - 1) + 2\delta b_{0,0} b_{0,2} - 12b_{2,0}^2 = 0. \quad (58c)$$

Furthermore, solving system of equation Eq. (58), we derive parametric values as

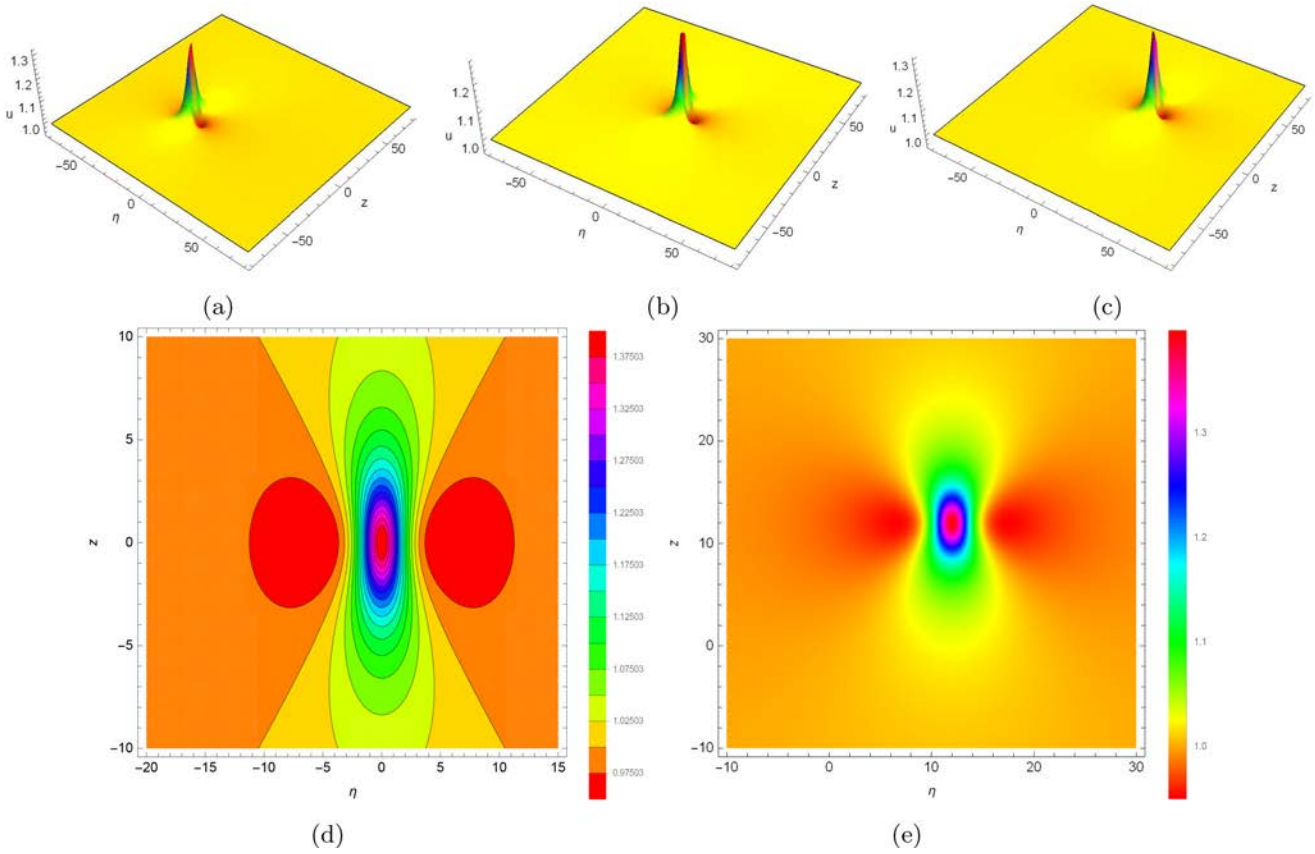


FIG. 7. First order rogue wave of Eq. (2) given by Eq. (61) with center controlling parameters as (a) $c_1 = -20$, $c_2 = -16$, (b) $c_1 = 0$, $c_2 = 0$, (c) $c_1 = 12$, $c_2 = 12$, (d) contour plot, and (e) density plot.

$$\begin{aligned} b_{2,0} &= b_{2,0}, & b_{0,2} &= \frac{b_{2,0}(a + \alpha + d - 1)}{\delta}, \\ b_{0,0} &= \frac{3b_{2,0}}{a + \alpha + d - 1}. \end{aligned} \quad (59)$$

With the parametric values in Eq. (59), Eq. (57) becomes

$$\begin{aligned} g(\rho, z) &= \tilde{g}_1(\rho, z, c_1, c_2) \\ &= b_{2,0} \left(\frac{(a + \alpha + d - 1)}{\delta} (z - c_2)^2 + (\rho - c_1)^2 + \frac{3}{a + \alpha + d - 1} \right). \end{aligned} \quad (60)$$

Finally, substitution of Eq. (60) into Eq. (54) yields first order rogue wave of Eq. (2) as

$$\begin{aligned} u(\rho, z) &= -\frac{6}{b} \left[\log \left(b_{2,0} \left(\frac{(a + \alpha + d - 1)}{\delta} (z - c_2)^2 + (\rho - c_1)^2 \right. \right. \right. \\ &\quad \left. \left. \left. + \frac{3}{a + \alpha + d - 1} \right) \right) \right]_{\rho\rho}. \end{aligned} \quad (61)$$

In Fig. 7, we demonstrate evolution of rogue wave solution of Eq. (2) corresponding to the parametric values $\alpha = 0.5$, $\delta = 0.3$, $a = 0.5$, $b = 1$, $c = -1$, $d = 0.3$, and $b_{2,0} = 1$.

VIII. LUMP SOLUTION

The lump solution^{74,85} of Eq. (2) can be constructed by choosing the test function in the following form:

$$f = h_1^2 + h_2^2 + m_1, \quad (62)$$

where

$$h_1 = d_1x + d_2y + d_3z + d_4t + d_5, \quad (63a)$$

$$h_2 = d_6x + d_7y + d_8z + d_9y + d_{10}, \quad (63b)$$

where m_1 , d_i , ($i = 1, 2, \dots, 10$) are constants that will be determined later, and the constants d_i ($i = 1, 2, 3, 6, 7, 8$) satisfy the condition $(d_1, d_2, d_3) \nparallel (d_6, d_7, d_8)$. Substituting Eq. (61) into Eq. (20) and setting all the coefficient of different powers of $x^p y^q z^r t^s$ to zero, we obtain a system of equations. After solving the obtained system of equations with $\gamma = 1$, $\delta = \frac{1}{4d}$, we derive the following restriction between the parameters:

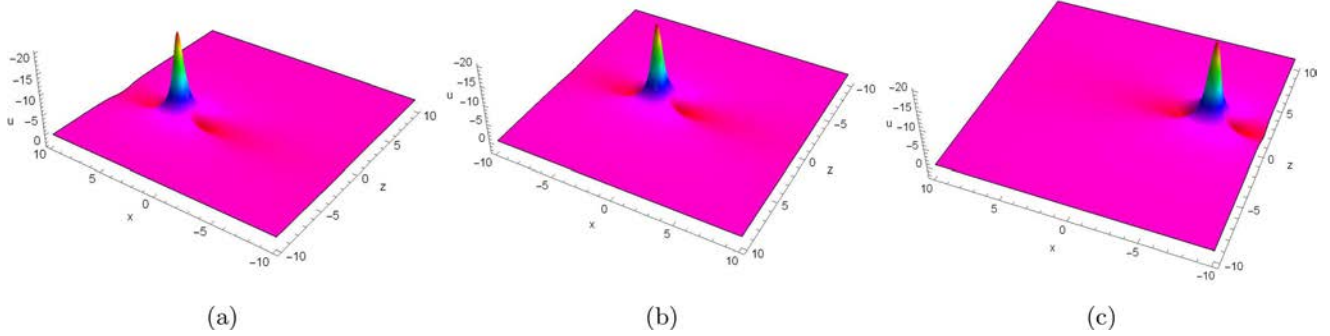


FIG. 8. Evolution of negative lump wave of Eq. (2) at (a) $t = -1.5$, (b) $t = 0$, and (c) $t = 1$.

$$d_2 = \frac{d_6 d_7}{d_1}, \quad d_3 = -\frac{d_6 d_8}{d_1}, \quad d_4 = \frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1},$$

$$m_1 = -\frac{12d c d_1^2 (d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (64a)$$

$$d_9 = -\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2},$$

$$d_{10} = \frac{d_5 d_6}{d_1}, \quad d_8 \neq -2d d_7, \quad c < 0. \quad (64b)$$

Finally, substituting Eq. (61) along with Eq. (64) into Eq. (20), we can derive lump solution of Eq. (2).

In Fig. 8, we demonstrate evolution of negative lump wave solution of Eq. (2) corresponding to the parametric values $a = b = d = \alpha = \beta = 1$, $c = -2$, $d_1 = 1$, $d_5 = 2$, $d_6 = 2$, $d_7 = 1$, and $d_8 = 2.5$.

IX. LUMP-MULTI-STRIPE SOLUTION

To construct lump-multi-stripe solution from Hirota bilinear form Eq. (20) of Eq. (2), we can consider the test function f in a combined form of quadratic function and exponential function as

$$f = h_1^2 + h_2^2 + m_1 + \sum_{i=1}^n e^{Y_i}, \quad (65)$$

where $Y_i = k_i(x + p_i y + q_i z + w_i t) + Y_i^0$, ($i = 1, 2, 3, n$), $m_1 > 0$ and h_1 and h_2 are defined in Eq. (63).

The test function f in Eq. (65) can provide lump-multi-stripe solution⁸¹ of Eq. (2) if and only if

$$(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2)(h_1^2 + h_2^2 + m_1) \cdot (h_1^2 + h_2^2 + m_1) = 0, \quad (66a)$$

$$(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2)(h_1^2 + h_2^2 + m_1) \cdot e^{Y_i}, \quad (i = 1, 2, \dots, n), \quad (66b)$$

$$\left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2 \right) (k_i - k_j, p_i - p_j, q_i - q_j, w_i - w_j) = 0, \quad (i \neq j, i, j = 1, 2, \dots, n). \quad (66c)$$

Setting $\gamma = 1$, $\delta = \frac{1}{4d}$ and solving system of equations Eq. (66), we derive relation between parameters $m_1, d_i, (i = 1, 2, \dots, 10)$ and k_i, p_i, q_i, w_i ($i = 1, 2, \dots, n$) as

$$d_2 = \frac{d_6 d_7}{d_1}, \quad d_3 = -\frac{d_6 d_8}{d_1}, \quad d_4 = \frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1}, \quad m_1 = -\frac{12d c d_1^2 (d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (67a)$$

$$d_9 = -\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2}, \quad d_{10} = \frac{d_5 d_6}{d_1}, \quad d_8 \neq -2d d_7, \quad c < 0, \quad (67b)$$

$$w_i = -\left(a + c k_i^2 + d p_i^2 + \alpha p_i + \beta q_i + p_i q_i + \frac{1}{4d} q_i^2 \right), \quad (i = 1, 2, \dots, n). \quad (67c)$$

Then, the lump-multi-stripe solution of Eq. (2) is obtained as

$$u = \frac{6c}{b} \left[\log \left(h_1^2 + h_2^2 + m_1 + \sum_{i=1}^n e^{\Upsilon_i} \right) \right]_{xx}, \quad (68)$$

where

$$h_1 = d_1 x + \left(\frac{d_6 d_7}{d_1} \right) y + \left(-\frac{d_6 d_8}{d_1} \right) z + \left(\frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1} \right) t + d_5, \quad (69a)$$

$$h_2 = d_6 x + d_7 y + d_8 z + \left(-\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2} \right) t + \frac{d_5 d_6}{d_1}, \quad (69b)$$

$$m_1 = -\frac{12d c d_1^2(d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (69c)$$

$$\Upsilon_i = k_i \left(x + p_i y + q_i z - \left(a + c k_i^2 + d p_i^2 + \alpha p_i + \beta q_i + p_i q_i + \frac{1}{4d} q_i^2 \right) t \right) + \Upsilon_i^0, \quad (i = 1, 2, \dots, n). \quad (69d)$$

A. Lump-one-stripe solution

For lump-one-stripe solution of Eq. (2), we take $i = 1$ in Eq. (68) and obtain lump-one-stripe solution as

$$u = \frac{6c}{b} \left[\log \left(h_1^2 + h_2^2 + m_1 + \sum_{i=1}^n e^{\Upsilon_i} \right) \right]_{xx}, \quad (70)$$

where

$$h_1 = d_1 x + \left(\frac{d_6 d_7}{d_1} \right) y + \left(-\frac{d_6 d_8}{d_1} \right) z + \left(\frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1} \right) t + d_5, \quad (71a)$$

$$h_2 = d_6 x + d_7 y + d_8 z + \left(-\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2} \right) t + \frac{d_5 d_6}{d_1}, \quad (71b)$$

$$m_1 = -\frac{12d c d_1^2(d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (71c)$$

$$\Upsilon_1 = k_1 \left(x + p_1 y + q_1 z - \left(a + c k_1^2 + d p_1^2 + \alpha p_1 + \beta q_1 + p_1 q_1 + \frac{1}{4d} q_1^2 \right) t \right) + \Upsilon_1^0. \quad (71d)$$

In Fig. 9, we demonstrate interaction phenomena of negative lump wave with stripe wave and their evolution at different time frames corresponding to the parametric values as $a = b = d = \alpha = \beta = 1$, $c = -1$, $d_1 = 1$, $d_5 = 1$, $d_6 = 2$, $d_7 = 1$, $d_8 = 1.2$, $k_1 = 2$, $p_1 = 2$, $q_1 = 0.25$, and $\Upsilon_1^0 = 0$. At $t = -1.5$, there is only one stripe wave. Lump wave originates from stripe wave at $t = 0$. Then, at $t = 0.5$, lump wave separates from stripe wave and move on.

B. Lump-two-stripe solution

For lump-two-stripe solution of Eq. (2), we take $i = 2$ in Eq. (68) and obtain lump-two-stripe solution as

$$u = \frac{6c}{b} \left[\log \left(h_1^2 + h_2^2 + m_1 + e^{\Upsilon_1} + e^{\Upsilon_2} \right) \right]_{xx}, \quad (72)$$

where

$$h_1 = d_1 x + \left(\frac{d_6 d_7}{d_1} \right) y + \left(-\frac{d_6 d_8}{d_1} \right) z + \left(\frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1} \right) t + d_5, \quad (73a)$$

$$h_2 = d_6 x + d_7 y + d_8 z + \left(-\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2} \right) t + \frac{d_5 d_6}{d_1}, \quad (73b)$$

$$m_1 = -\frac{12d c d_1^2(d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (73c)$$

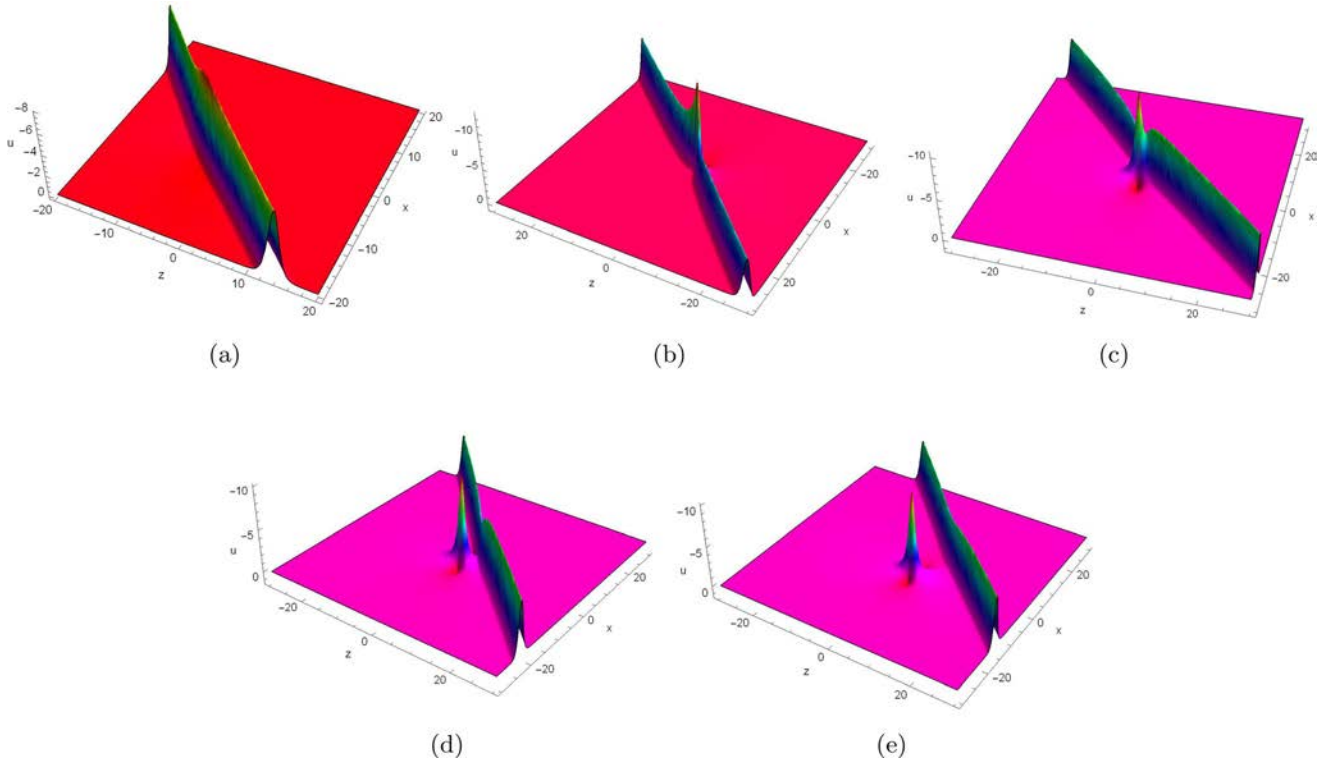


FIG. 9. Evolution of negative lump-one-stripe solution of Eq. (2) at (a) $t = -1.5$, (b) $t = -0.5$, (c) $t = 0$, (d) $t = 0.5$, and (e) $t = 1.5$.

$$\Upsilon_i = k_i \left(x + p_i y + q_i z - \left(a + c k_i^2 + d p_i^2 + \alpha p_i + \beta q_i + p_i q_i + \frac{1}{4d} q_i^2 \right) t \right) + \Upsilon_i^0, \quad (i = 1, 2). \quad (73d)$$

In Fig. 10, we demonstrate evolution of lump-two-stripe solution of Eq. (2) corresponding to the parametric values as $a = b = d = \alpha = \beta = 1$, $c = -1$, $d_1 = 1$, $d_5 = 1$, $d_6 = 2$, $d_7 = 1$, $d_8 = 1.2$, $k_1 = 2$, $p_1 = 2$, $q_1 = 0.25$, $k_2 = 1$, $p_2 = 1$, $q_2 = 0.25$, and $\Upsilon_i^0 = 0$, ($i = 1, 2$). At $t = -1.5$, there is only one stripe wave, and lump wave originates from stripe wave at $t = 0$. Lump wave completely isolates from stripe wave at $t = 2$. At $t = 3.5$, stripe wave splits into two distinct stripe waves and then move on.

X. LUMP-MULTI-SOLITON SOLUTION

To construct a lump-multi-soliton solution from Hirota bilinear form Eq. (20) of Eq. (2), we can consider the test function f in a combined form of quadratic function and hyperbolic cosine function as

$$f = h_1^2 + h_2^2 + m_1 + \sum_{i=1}^n \cosh \Upsilon_i + m_2, \quad (74)$$

where $\Upsilon_i = k_i(x + p_i y + q_i z + w_i t) + \Upsilon_i^0$, ($i = 1, 2, 3, n$), $m_1, m_2 > 0$, and h_1 and h_2 are defined in Eq. (63).

The test function f in Eq. (74) can provide lump-multi-soliton solution⁸¹ of Eq. (2) if and only if

$$\left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2 \right) (h_1^2 + h_2^2 + m_1) \cdot (h_1^2 + h_2^2 + m_1) = 0, \quad (75a)$$

$$\left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2 \right) (h_1^2 + h_2^2 + m_1) \cdot e^{\Upsilon_i}, \quad (i = 1, 2, \dots, n), \quad (75b)$$

$$\begin{aligned} & \left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2 \right) (h_1^2 + h_2^2 + m_1) \cdot m_2 \\ & + \sum_{i=1}^n \frac{1}{4} \left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2 \right) (2k_i, 2p_i, 2q_i, 2w_i) = 0, \end{aligned} \quad (75c)$$

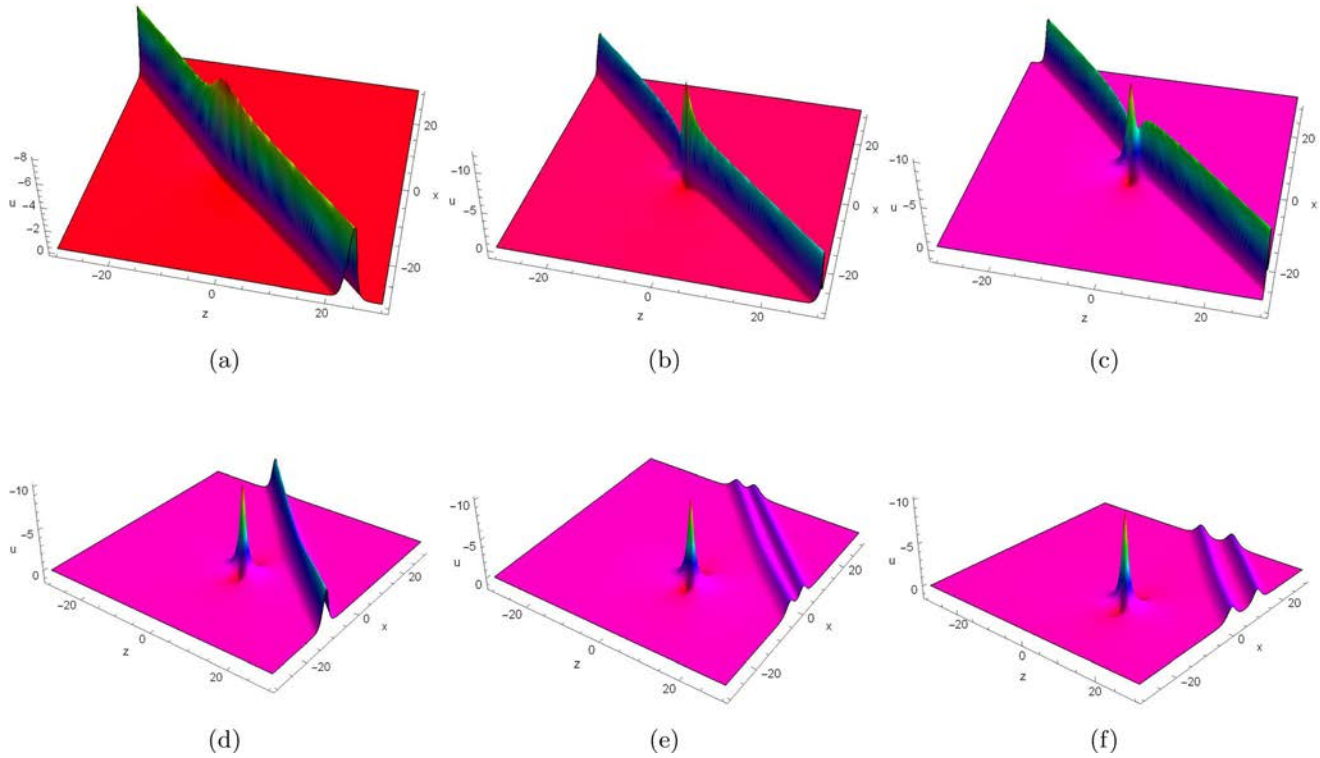


FIG. 10. Evolution of negative lump-two-stripe solution of Eq. (2) at (a) $t = -1.5$, (b) $t = -0.5$, (c) $t = 0$, (d) $t = 2$, (e) $t = 3.5$, and (f) $t = 4.5$.

$$\left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2\right)(k_i - k_j, p_i - p_j, q_i - q_j, w_i - w_j) = 0, \quad (75d)$$

$$(i \neq j, i, j = 1, 2, \dots, n),$$

$$\left(D_x D_t + a D_x^2 + c D_x^4 + d D_y^2 + \alpha D_x D_y + \beta D_x D_z + \gamma D_y D_z + \delta D_z^2\right)(k_i + k_j, p_i + p_j, q_i + q_j, w_i + w_j) = 0, \quad (75e)$$

$$(i \neq j, i, j = 1, 2, \dots, n).$$

Setting $\gamma = 1$, $\delta = \frac{1}{4d}$ and solving system of equations Eq. (75), we derive same relations between parameters m_1 , d_i , ($i = 1, 2, \dots, 10$) and k_i, p_i, q_i, w_i ($i = 1, 2, \dots, n$) as in Eq. (67).

Then, the lump-multi-soliton solution of Eq. (2) is obtained as

$$u = \frac{6c}{b} \left[\log \left(h_1^2 + h_2^2 + m_1 + \sum_{i=1}^n \cosh \Upsilon_i + m_2 \right) \right]_{xx}, \quad (76)$$

where

$$h_1 = d_1 x + \left(\frac{d_6 d_7}{d_1} \right) y + \left(-\frac{d_6 d_8}{d_1} \right) z + \left(\frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1} \right) t + d_5, \quad (77a)$$

$$h_2 = d_6 x + d_7 y + d_8 z + \left(-\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2} \right) t + \frac{d_5 d_6}{d_1}, \quad (77b)$$

$$m_1 = -\frac{12d c d_1^2(d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (77c)$$

$$\Upsilon_i = k_i \left(x + p_i y + q_i z - \left(a + c k_i^2 + d p_i^2 + \alpha p_i + \beta q_i + p_i q_i + \frac{1}{4d} q_i^2 \right) t \right) + \Upsilon_i^0, \quad (i = 1, 2, \dots, n). \quad (77d)$$

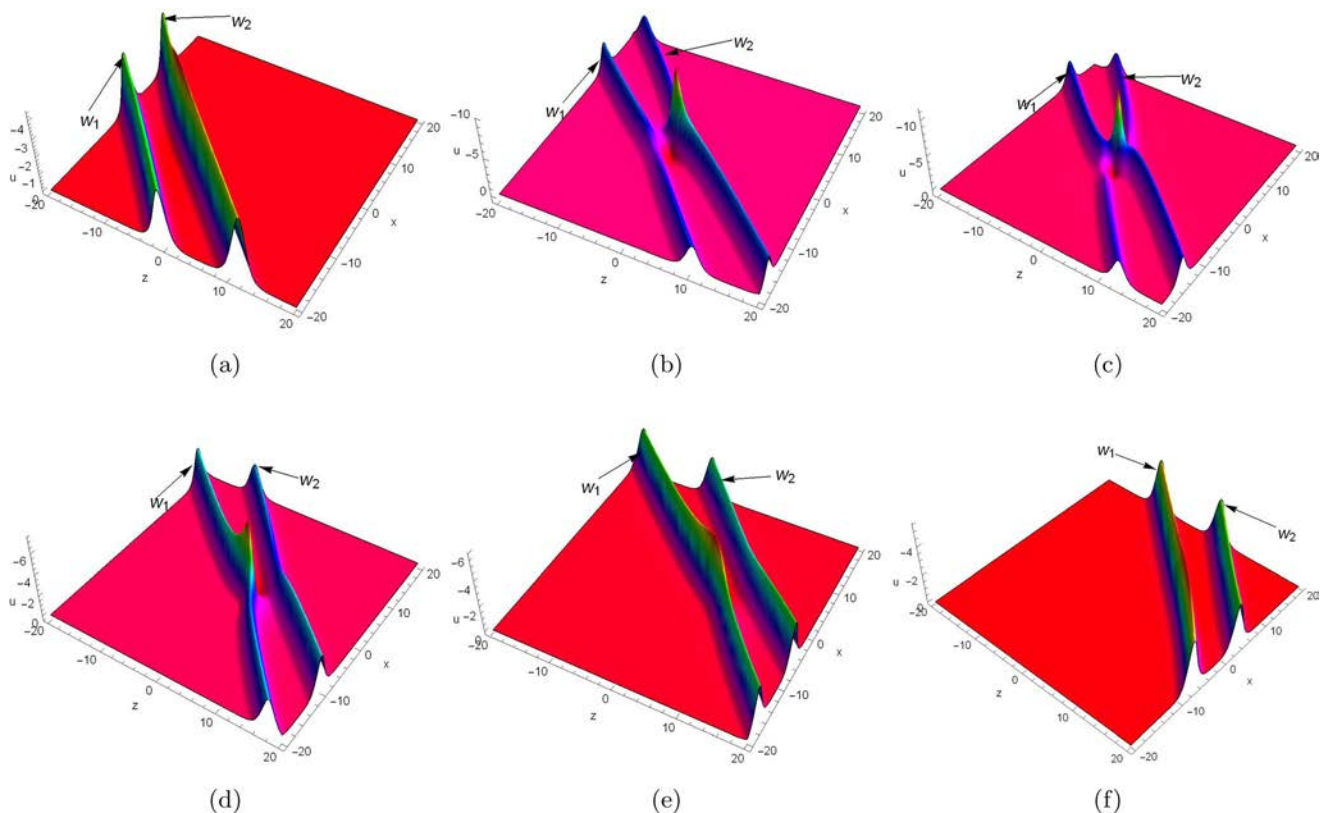


FIG. 11. Evolution of negative lump-one-soliton solution of Eq. (2) given by Eq. (78) at (a) $t = -4$, (b) $t = -1$, (c) $t = -0.3$, (d) $t = 1$, (e) $t = 2.3$, and (f) $t = 4.8$.

A. Lump-one-soliton solution

For lump-one-soliton solution of Eq. (2), we take $i = 1$ in Eq. (76) and obtain lump-one-soliton solution as

$$u = \frac{6c}{b} [\log(h_1^2 + h_2^2 + m_1 + \cosh \Upsilon_1 + m_2)]_{xx}, \quad (78)$$

where

$$h_1 = d_1 x + \left(\frac{d_6 d_7}{d_1}\right) y + \left(-\frac{d_6 d_8}{d_1}\right) z + \left(\frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1}\right) t + d_5, \quad (79a)$$

$$h_2 = d_6 x + d_7 y + d_8 z + \left(-\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2}\right) t + \frac{d_5 d_6}{d_1}, \quad (79b)$$

$$m_1 = -\frac{12d c d_1^2(d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (79c)$$

$$\Upsilon_1 = k_1 \left(x + p_1 y + q_1 z - \left(a + c k_1^2 + d p_1^2 + \alpha p_1 + \beta q_1 + p_1 q_1 + \frac{1}{4d} q_1^2 \right) t \right) + \Upsilon_1^0. \quad (79d)$$

In Fig. 11, we illustrate the evolution of negative lump-one-soliton solution of Eq. (2) at different time frames corresponding to the parametric values as $a = b = d = \alpha = \beta = 1$, $c = -1$, $d_1 = 1$, $d_5 = 1$, $d_6 = 2$, $d_7 = 1$, $d_8 = 1.3$, $k_1 = 1.5$, $p_1 = 1$, $q_1 = 0.25$, $m_2 = 0$, and $\Upsilon_1^0 = 0$. At $t = -4$, there are two solitary waves w_1 and w_2 . Lump wave originates from w_2 at $t = -1$ and then disappears into wave w_1 with time.

B. Lump-two-soliton solution

For lump-two-soliton solution of Eq. (2), we take $i = 2$ in Eq. (76) and obtain lump-two-soliton solution as

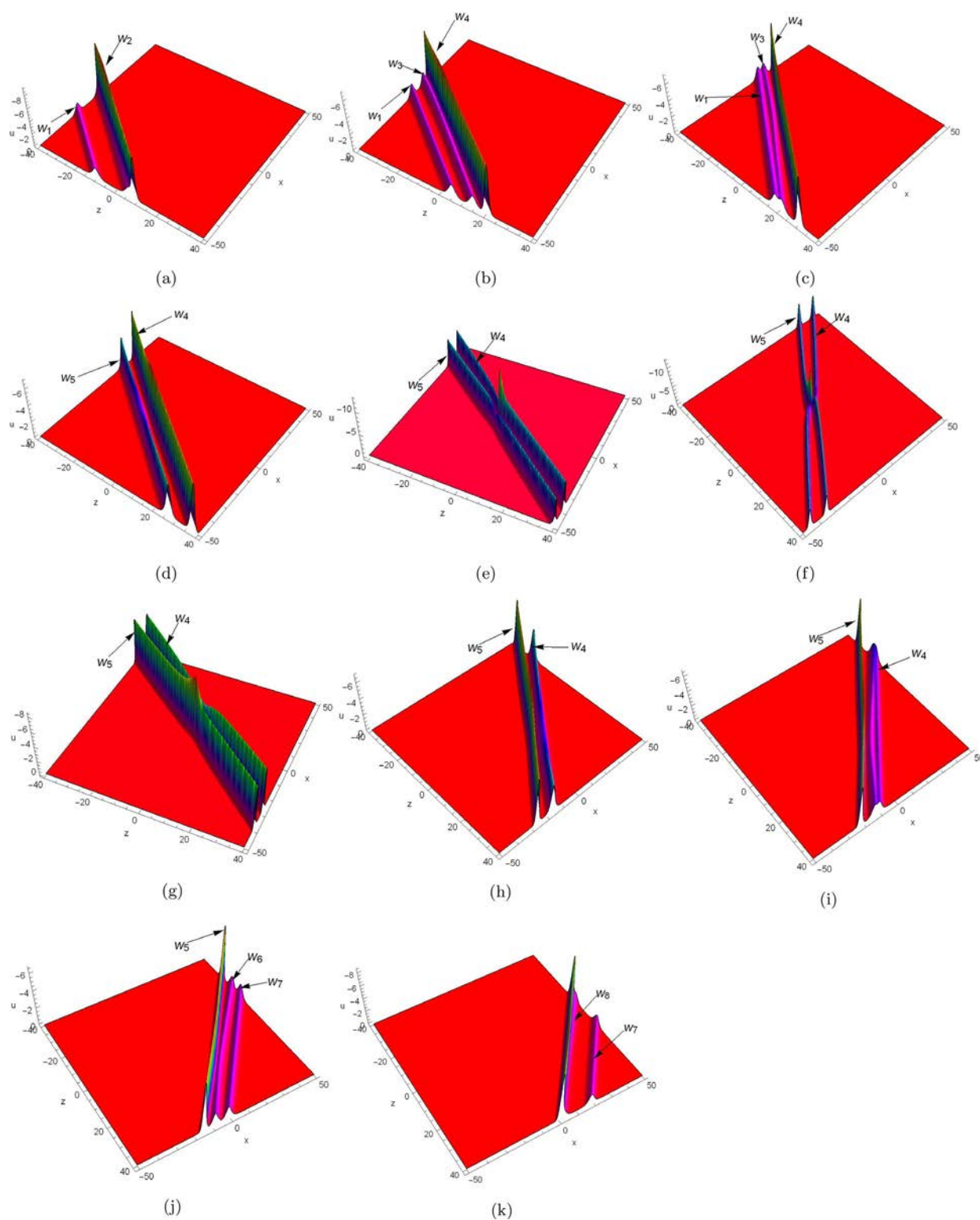


FIG. 12. Evolution of negative lump-two-soliton solution of Eq. (2) given by Eq. (80) at (a) $t = -6.7$, (b) $t = -5$, (c) $t = -3.6$, (d) $t = -2.6$, (e) $t = -0.5$, (f) $t = 0$, (g) $t = 1$, (h) $t = 2.7$, (i) $t = 3.3$, (j) $t = 4.3$, and (k) $t = 6.5$.

$$u = \frac{6c}{b} [\log(h_1^2 + h_2^2 + m_1 + \cosh \Upsilon_1 + \cosh \Upsilon_2 + m_2)]_{xx}, \quad (80)$$

where

$$h_1 = d_1 x + \left(\frac{d_6 d_7}{d_1}\right) y + \left(-\frac{d_6 d_8}{d_1}\right) z + \left(\frac{4dd_6(\alpha d_7 + \beta d_8) - 4d(ad_1^2 - d d_7^2) + 4d d_8 d_7 + d_8^2}{4d d_1}\right) t + d_5, \quad (81a)$$

$$h_2 = d_6 x + d_7 y + d_8 z + \left(-\frac{4d d_1^2(\alpha d_7 + \beta d_8) + 4d d_6(ad_1^2 - d d_7^2) - (4d d_6 d_7 d_8 + d_6 d_8^2)}{d_1^2}\right) t + \frac{d_5 d_6}{d_1}, \quad (81b)$$

$$m_1 = -\frac{12d c d_1^2 (d_1^2 + d_6^2)}{(2d d_7 + d_8)^2}, \quad (81c)$$

$$\Upsilon_i = k_i \left(x + p_i y + q_i z - \left(a + c k_i^2 + d p_i^2 + \alpha p_i + \beta q_i + p_i q_i + \frac{1}{4d} q_i^2 \right) t \right) + \Upsilon_i^0, \quad (i = 1, 2). \quad (81d)$$

In Fig. 12, we illustrate the evolution of negative lump-two-soliton solution of Eq. (2) corresponding to the parametric values as $a = b = d = \alpha = \beta = 1$, $c = -1$, $d_1 = 1$, $d_5 = 1$, $d_6 = 2$, $d_7 = 1$, $d_8 = 1.3$, $k_1 = 1.5$, $p_1 = 2$, $q_1 = 0.25$, $k_2 = 1$, $p_2 = 1$, $q_2 = 0.25$, $m_2 = 0$, and $\Upsilon_i^0 = 0$ ($i = 1, 2$). Evolution of lump-two-soliton solution is more sophisticated due to the presence of various fissions and fusions of waves. At $t = -6.7$, exactly two waves w_1 and w_2 are appeared. Then, at $t = -5$, wave w_2 starts splitting to w_3 and w_4 , and at $t = -3.6$, it completely splits because of the fission effect. At $t = -2.6$, wave w_5 appears as a result of the fusion effect of waves w_1 and w_3 . At $t = -0.5$, lump wave originates from w_4 and completely disappears into wave w_5 at $t = 2.7$. Then, wave w_4 splits into w_6 and w_7 . Finally, because of the fusion effect of w_5 and w_6 , wave w_8 generates.

XI. GRAPHICAL ILLUSTRATION, RESULTS, AND DISCUSSION

In Fig. 1, we depict 3D plot, density plot, and 2D plot of one soliton solution of Eq. (2) corresponding to the parametric values as $a = c = d = \alpha = \beta = \gamma = \delta = 1$, $k_1 = 0.4$, $p_1 = 1.2$, $q_1 = 1.3$, and $\Upsilon_1^0 = 0$. Similarly, in Figs. 2 and 4, we present a 3D plot, density plot, and 2D plot of two soliton solutions and three soliton solution of Eq. (2), choosing parametric values as $a = c = d = \alpha = \beta = \gamma = \delta = 1$, $k_1 = 0.9$, $p_1 = 1.2$, $q_1 = 1.3$, $k_2 = -1.3$, $p_2 = 0.7$, $q_2 = 2$, and $\Upsilon_i^0 = 0$ ($i = 1, 2$) and $a = c = d = \alpha = \beta = \gamma = \delta = 1$, $k_1 = 0.9$, $p_1 = 1.2$, $q_1 = 1.3$, $k_2 = -1.3$, $p_2 = 0.7$, $q_2 = 2$, $k_3 = 1.7$, $p_3 = 1$, $q_3 = 1.7$, and $\Upsilon_i^0 = 0$ ($i = 1, 2, 3$), respectively. Figures 3 and 5 show the propagation of two soliton solution and three soliton solution of the model equation. Notably, larger solitary waves collide elastically with smaller waves, and after the nonlinear interaction, all waves retain their shapes asymptotically. The larger waves continue moving ahead of the smaller ones, demonstrating that the soliton solutions maintain their original shape and size with only a phase change during interaction. In Fig. 6, we illustrate the dynamical behavior of the solution given by Eq. (52) at different time frames. At the outset, waves appear from a constant state and progressively transform into periodic line waves. These periodic line waves maintain parallelism and independence from each other, yet their behaviors evolve consistently over time. In Fig. 7, we demonstrate 3D plots of first order rogue wave solution corresponding to the different center controlling parameters as $c_1 = -20$, $c_2 = -16$; $c_1 = 0$, $c_2 = 0$; and $c_1 = 12$, $c_2 = 12$.

In Fig. 8, we present evolution of lump solution corresponding to the parametric values as $a = b = d = \alpha = \beta = 1$, $c = -2$, $d_1 = 1$, $d_5 = 2$, $d_6 = 2$, $d_7 = 1$, and $d_8 = 2.5$. Figure 9 depicts the evolutionary process of a lump solution emerging from a stripe solution. In the lump-one-stripe solution, a lump wave generates from a stripe wave and gradually separates from the stripe wave over time. Figure 10 illustrates the evolutionary process and fission effect of lump and stripe solutions. In the lump-two-stripe solution, a lump wave originates from the stripe wave, and subsequently, the lump and stripe waves separate after a certain duration. Following this separation, the stripe wave further divides into two distinct stripe waves. In Fig. 11, we illustrate the evolutionary process of a lump solution arising from a soliton solution, along with their interaction dynamics. In the lump-one-soliton solution, a lump wave emanates from one of the solitary waves, gradually detaches from that solitary wave and interacts with another solitary wave. After that, lump wave fades away entirely and transforms into another solitary wave over time. This indicates that the interaction between lump and one-soliton is entirely inelastic. Figure 12 depicts the occurrence of multiple fission and fusion effects in the evolution of a lump solution from soliton solutions. Within the lump-two-soliton solution, various instances of fission and fusion phenomena occur. Initially, there are only two solitary waves, denoted as w_1 and w_2 . Subsequently, w_2 divides into two solitary waves, w_3 and w_4 . Following this, w_3 interacts with w_1 , giving rise to the formation of a new solitary wave, w_5 . A lump wave originates from w_4 , gradually separates from it, interacts with w_5 , and eventually dissipates completely within w_5 . Following this, w_5 splits into two solitary waves, w_6 and w_7 . Again, w_6 interacts with w_5 , resulting in the emergence of a new solitary wave, w_8 . This suggests that the interaction between the lump and the two-soliton is also completely nonelastic.

XII. CONCLUSION

In this article, we have delved into the intricacies of the integrability characteristics inherent in a novel $(3+1)$ -dimensional nonlinear evolution equation. Our approach commences with the elucidation of fundamental properties rooted in Bell polynomial theory. Leveraging this theoretical framework, we derive both the Hirota bilinear form and a bilinear Bäcklund transformation. By employing the Cole–Hopf transformation within the bilinear Bäcklund transformation and subsequently linearizing it, we arrive at the formulation of a Lax pair.

Furthermore, our exploration extends to the integrability aspect of the model equation, involving the derivation of an infinite sequence of conservation laws. The analytical effectiveness of the Hirota bilinear form facilitates the derivation of one, two, and three soliton solutions, vividly portrayed through graphical representations. Beyond solitons, we delve into the realm of breather solutions, introducing complex conjugate parameters into the two-soliton solution and visually depicting their evolutionary patterns. Our exploration takes an intriguing turn as we derive a first-order center-controlled rogue wave solution, dynamically illustrating its behavior across varying center-controlling parameter values. Expanding the scope of our inquiry, we employ the well-established quadratic function as a test function within the Hirota bilinear form, resulting in the deduction of the lump solution. Furthermore, a novel approach utilizing a combined form of a quadratic function and an exponential function as a test function yields lump-multi-stripe solutions, capturing their evolutionary phenomena through graphical representations. Pushing the boundaries of our investigation, we delve into the synthesis of lump-multi-soliton solutions through the combined application of a quadratic function and hyperbolic cosine function as test functions. The visually compelling depictions of their evolutionary phenomena add another layer to our understanding of these complex nonlinear wave dynamics. Significantly, these discoveries enhance our understanding of nonlinear wave phenomena across diverse fields, including shallow water dynamics, oceanography, and nonlinear optics. The results derived from this analytical framework offer a valuable and precise method for interpreting and describing various intricate phenomena within these scientific domains. In future endeavors, our goal is to apply the linear superposition principle to the Hirota bilinear form within a complex field. This approach seeks to extract solutions in the form of complex exponential wave functions, giving rise to phenomena such as complexitons, resonant solitons, and related effects. Additionally, another avenue of our research will involve exploring symmetry reductions of exact solutions through Lie symmetry analysis.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Uttam Kumar Mandal: Conceptualization (equal); Methodology (equal); Software (equal); Writing—original draft (equal). **Amiya Das:** Formal analysis (lead); Resources (equal); Software (equal); Supervision

(lead); Validation (lead); Visualization (lead); Writing—review & editing (lead). **Wen-Xiu Ma:** Formal analysis (equal); Supervision (equal); Visualization (equal); Writing—review & editing (equal).

DATA AVAILABILITY

All data generated or analyzed during this study are included in this article.

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